n8n AI Agent Development & Implementation Program

Master Engineering Handbook: Practical Step-by-Step Implementation from Zero to Autonomous Multi-Agent Systems

Document Type: Master Corporate Technical Training Handbook & Assignment Workbook
Primary Platform: n8n Automation Engine (v1.60+) with LangChain Cluster Nodes
Foundational AI Models: Google Gemini 1.5 Flash / Pro (Chat) & text-embedding-004 (Embeddings)
Vector Database: Pinecone Serverless (768 Dimensions, Cosine Metric)
Protocols Covered: ReAct Reasoning, Vector RAG, Model Context Protocol (MCP), Agent-to-Agent (A2A)
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Edition: Enterprise Production Edition 2026

EXECUTIVE TRAINING BRIEFING

This master document provides complete step-by-step guidance for corporate training sessions. It includes foundational theory, live trainer demonstration scripts, a library of 22 simple, practical business use cases with exact configurations, testing matrices, and evaluation rubrics.

Comprehensive Training Program Roadmap

The curriculum is structured into 12 rigorous modules, providing an unbroken progression from baseline automation to autonomous agent swarms:

Curriculum Section	Core Competencies Delivered
Module 1: AI & LLM Fundamentals from Scratch	Tokens, tokenization, context windows, temperature, hallucinations, embeddings, and ReAct.
Module 2: n8n Automation Engine & Core Architecture	Node graph mechanics, execution history, expressions ($json, nodeJson), and data mapping.
Module 3: Pre-Workshop Environment Setup	Turnkey registration & API verification for n8n, Google Gemini, and Pinecone Serverless.
Module 4: Production Prompt Engineering	5-pillar prompt architecture, structured JSON schemas, and defensive prompt injection filtering.
Module 5: Designing Autonomous AI Agents in n8n	AI Agent nodes, subnodes (Model, Memory, Tools, Parsers), and Trainer Live Demo 1 (Support Triage).
Part I: Foundational Agent Assignments	12 complete, simple business use cases with 16-point specifications for individual participants.
Module 6: Vector Databases & Embeddings Architecture	Semantic vector space, chunking strategies, overlap rules, and Pinecone index management.
Module 7: Retrieval-Augmented Generation (RAG) Mastery	Offline ingestion vs online retrieval, grounding, and Trainer Live Demo 2 (Company Policy RAG).
Part II: Semantic Retrieval (RAG) Assignments	10 domain-specific RAG projects with knowledge bases, metadata schemas, and factuality checklists.
Module 8: Advanced Agent Capabilities & Memory	Heterogeneous tools (HTTP, Code, DB), session memory, and Human-in-the-Loop approval nodes.
Module 9: Model Context Protocol (MCP) Integration	MCP client-server architecture, exposing n8n workflows as tools, consuming external servers.
Module 10: Agent-to-Agent (A2A) Orchestration	Hierarchical coordinator-specialist architectures, task delegation, and multi-agent systems.
Module 11: Production Readiness & Governance	Credential vaulting, rate limiting, error alert workflows, evaluation datasets, and audit trails.
Module 12: Final Capstone Project & Evaluation	6 enterprise project tracks, submission requirements, 10-minute live demo, and 100-point rubric.
Master Facilitator Session Timeline	Detailed minute-by-minute agenda for the full-day Saturday intensive workshop.
Appendix: Master Glossary of AI & n8n Terminology	Authoritative definitions for all critical agentic, RAG, and automation terms.

Module 1: AI & LLM Fundamentals from Scratch

1.1 The AI Hierarchy: From Rules to Generative Models

To engineer effective agentic workflows, developers must understand that Large Language Models (LLMs) are not databases or deterministic calculation engines. They are **probabilistic next-token predictors**. When an LLM receives an input prompt, it computes probability distributions across its learned vocabulary to generate subsequent tokens.

Core Concept	Technical Definition	Practical Implication for n8n Developers
Token	Atomic linguistic chunk (~4 characters or 0.75 words).	API costs and context limits are strictly measured in tokens.
Context Window	Total token capacity processed in a single inference call.	Gemini 1.5 Flash provides 1M tokens; ideal for massive documents.
Temperature	Hyperparameter (0.0 to 1.0+) controlling sampling randomness.	Set to 0.0 for strict JSON extraction and classification; 0.7 for creative drafts.
Hallucination	Plausible-sounding output unsupported by factual data.	Must be mitigated using Grounding and Vector RAG context injection.
Embedding	High-dimensional vector array representing semantic meaning.	Enables mathematical similarity search across documents in Pinecone.

1.2 The Evolution: LLM vs. Assistant vs. Agent vs. Agentic AI

Dimension	Raw LLM	AI Assistant (Chatbot)	Autonomous AI Agent	Agentic AI System
Control Flow	Stateless, single turn	Multi-turn, user-guided	Goal-oriented ReAct loop	Multi-agent coordination
Tool Usage	None (text only)	Static pre-set plugins	Autonomous API & DB calls	Dynamic cross-agent tools
Memory	Zero internal state	Ephemeral chat window	Buffer + Vector memory	Distributed shared state
Autonomy	Passive text generation	Waits for user prompt	Decomposes goals, handles errors	Self-healing enterprise mesh

1.3 The LLM Request Lifecycle

Understanding how a prompt flows from user input to final completion prevents architectural errors:

1. **User Input:** Raw text submitted via Webhook or Chat.
2. **Prompt Construction:** System Prompt + Injected Context + Conversation Memory + User Message.
3. **Tokenization:** Text is converted into token integer IDs.
4. **Transformer Inference:** Attention heads evaluate relationships; softmax layer generates probabilities.
5. **De-Tokenization & Streaming:** Highest probability tokens converted back to text and returned to n8n.

Module 2: n8n Automation Engine & Core Architecture

2.1 Why n8n is the Premier Low-Code Agent Canvas

Writing agent logic in pure code (e.g. raw Python scripts) forces teams to reinvent connection pooling, queuing, state serialization, error recovery, API authentication, and execution logging. n8n provides all of this out of the box with complete visual observability.

2.2 Dynamic Expressions and the Critical Subnode Scoping Rule

Every node in n8n consumes an array of JSON objects and outputs an array of JSON objects. Understanding how data traverses n8n is vital:

```
// 1. Standard node data access (Edit Fields, HTTP Request, IF):
{{ $json.myVariable }}                       // Current item field
{{ $('Webhook').first().json.body.email }}    // Upstream node reference

// 2. CRUCIAL RULE inside AI Agent Subnodes (Tools, Parsers, Memory):
// Subnodes DO NOT share item context with predecessor nodes!
// Never use $json inside a Tool subnode. Instead use nodeJson():
{{ nodeJson('Webhook', 'body.customerId') }}
```

CRITICAL SUBNODE SCOPING RULE

The #1 mistake beginners make in n8n AI workflows is referencing {{ $json.field }} inside a Tool subnode. Subnodes execute in an isolated execution scope. Always reference upstream triggers using nodeJson('TriggerNodeName', 'path.to.key').

2.3 Hands-on Exercise: Data Transformation Pipeline

Build a verification flow: Webhook -> Edit Fields -> IF -> Respond to Webhook.

Node Name	Node Type	Key Parameter / Expression	Expected Behavior
Customer Ingest	Webhook	Path: /customer-ingest, Method: POST	Receives raw customer payload
Normalize Data	Edit Fields (Set)	name: ={{ $json.name.toUpperCase() }} vip: ={{ $json.total >= 100 }}	Transforms data types and computes boolean flag
Check VIP	IF	Conditions: vip Equal true	Branches flow based on VIP status
Send Response	Respond to Webhook	Response Body: ={{ JSON.stringify($json) }}	Returns clean JSON to calling client

Module 3: Pre-Workshop Prerequisites & Setup

3.1 Service Setup & SLA Matrix

To ensure a seamless hands-on learning experience, all accounts and API credentials must be provisioned and tested prior to live lab sessions:

Resource / Service	Purpose	Required For	Cost Tier & Quotas	Verification Action
n8n Instance	Workflow & Agent Orchestration	Module 2 (Day 1)	Cloud Free Trial / Self-Hosted Docker	Access canvas UI & test manual execution.
Google Gemini API	Chat Model & Embeddings	Module 4 (Day 1)	Google AI Studio Free Tier (15 RPM)	Execute curl generateContent test.
Pinecone Vector DB	Semantic Vector Store for RAG	Module 6 (Day 2)	Free Starter Tier (1 index, 100k vectors)	Create index 'n8n-rag-knowledgebase' (768 dims).
Web Browser	Canvas UI & Management	Immediate	Free (Chrome / Edge / Firefox)	Disable aggressive ad-blockers on n8n domain.

3.2 Secure Credential Registration in n8n

Rule: Never paste raw API keys into workflow notes, prompt bodies, or code blocks.

1. Open n8n -> Navigate to **Credentials** on left menu -> Click **Add Credential**.
2. Add **Google Gemini API** -> Paste API Key -> Click Save (Verifies with green checkmark).
3. Add **Pinecone API** -> Paste API Key -> Click Save (Verifies connection).

Module 4: Production Prompt Engineering

4.1 The 5-Pillar Production Prompt Framework

Enterprise prompt engineering is about **deterministic behavioral containment**. Every production prompt must include these 5 structural blocks:

Structural Block	Core Purpose	Production Example
1. Role & Identity	Establishes persona, domain authority & scope	'You are an Enterprise Support Triage Officer at TechFlow.'
2. Primary Task	Clear, unambiguous statement of work	'Analyze the customer email and classify urgency based on business impact.'
3. Context & Variables	Injected reference data or document chunks	'Context from CRM: {{ nodeJson("Webhook", "body.crmData") }}'
4. Negative Constraints	Explicit rules on what the model MUST NOT do	'NEVER invent account details. If missing ID, request it politely.'
5. Output Schema	Strict formatting directives or JSON structure	'Output your decision adhering strictly to the JSON schema provided.'

4.2 Five Before-and-After Production Prompt Examples

Use Case	Naive Prompt (Fragile / Prone to Failure)	Production Prompt (Robust & Deterministic)
Ticket Triage	Tell me if this customer is angry and what they want.	You are a Triage Officer. Classify email into: [BILLING, BUG, ACCESS]. Rate sentiment: [FRUSTRATED, NEUTRAL, HAPPY]. If downtime mentioned, set priority P1_CRITICAL.
Invoice Parsing	Extract all the information from this invoice.	Extract vendorName, invoiceId (INV-XXXXX), totalAmount (float, 2 decimals), dueDate (YYYY-MM-DD). If missing, set null and manualReviewRequired: true.
Policy Inquiries	Answer the employee's question about vacation.	Answer using ONLY the provided HR Policy Context. If not explicitly in context, reply word-for-word: 'This policy is not covered in the official 2026 handbook.'
SQL Generation	Write SQL to find all sales last month.	Generate a single SELECT query on table 'orders'. Output ONLY raw SQL. NEVER generate INSERT, UPDATE, DELETE, or DROP. Append LIMIT 50.
Standup Summary	Summarize what the developers said today.	Transform raw updates into 3 sections: 1. Accomplished Yesterday, 2. Planned Today, 3. Blockers. Highlight critical blockers in bold red.

Module 5: Designing Autonomous AI Agents in n8n

5.1 Anatomy of an n8n AI Agent

An n8n AI Agent node (@n8n/n8n-nodes-langchain.agent) coordinates reasoning and action via specialized subnode connectors:

Subnode Connector	Supported n8n Node	Configuration Parameters	Functional Responsibility
Model	Google Gemini Chat Model	Model: gemini-1.5-flash, Temp: 0.0	Core cognitive engine that evaluates prompts and tool calls.
Memory	Simple Memory	Window Buffer: 5 messages	Stores recent dialogue turns so the agent remembers context.
Tools	Custom Code Tool / Pinecone	Tool Name, Strict Tool Description	External actions the agent can choose to execute.
Output Parser	Structured Output Parser	JSON Schema Definition	Forces model completion into strictly validated JSON.

5.2 Trainer Live Demonstration 1: Customer Support Triage Agent

In this live build, the trainer constructs an autonomous support triage agent on an empty canvas. The agent accepts raw user messages, uses a custom tool to look up customer tier in a database, and generates structured priority classifications.

```
// Custom Code Tool Implementation: lookup_customer_tier
const customerDb = {
  "CUST-101": { name: "Acme Corp", tier: "ENTERPRISE_VIP", balance: 0 },
  "CUST-102": { name: "Bob Smith", tier: "STANDARD", balance: 45.00 }
};
const id = query.trim().toUpperCase();
return JSON.stringify(customerDb[id] || { error: "Customer not found." });
```

```
# System Prompt for Support Triage Agent
You are the Senior Customer Support Triage Specialist at TechFlow.
Your objective is to inspect incoming inquiries, lookup the customer tier using the lookup_customer_tier too
l, and draft an empathetic summary.

Strict Rules:
1. Look for a Customer ID (format: CUST-XXX). If present, immediately invoke the tool.
2. If tier is ENTERPRISE_VIP, mark priority as P1_CRITICAL.
3. If no Customer ID is present, politely request it while acknowledging their inquiry.
4. Never invent customer account details.
```

Part I: Foundational AI Agent Assignments (12 Use Cases)

Each participant is assigned one distinct, practical business project. All projects follow a simple, step-by-step implementation pattern without RAG complexity:

Case Identifier	Project Title	Core Functional Deliverable
Case 01 (Part. 1)	Customer Support Ticket Triager	Classify urgency (P1/P2/P3), extract sentiment, output structured JSON.
Case 02 (Part. 2)	Expense Receipt Classifier	Parse merchant, float amount, date, and tax category (Meals/Travel).
Case 03 (Part. 3)	Meeting Action Item Extractor	Extract tabular checklist with Task, Assignee, Deadline, and Dependencies.
Case 04 (Part. 4)	Sales Lead Qualification Agent	Evaluate BANT criteria; score lead 1-100; branch with IF node (HOT vs COLD).
Case 05 (Part. 5)	Order Status Inquiry Agent	Extract order ID; call Custom Code Tool (mock orders); return delivery ETA.
Case 06 (Part. 6)	IT Password & Access Triager	Identify system (Okta/Jira/AWS); assess risk; supply self-service URL.
Case 07 (Part. 7)	Vendor Invoice Data Extractor	Extract line items, subtotal, tax, and verify arithmetic total.
Case 08 (Part. 8)	Candidate Resume Screener	Compare resume against Job Description; score match %; list strengths & gaps.
Case 09 (Part. 9)	Appointment Scheduling Coordinator	Normalize relative dates to ISO; verify working hours; call slot tool.
Case 10 (Part. 10)	Daily Standup Summarizer	Consolidate fragmented team updates into Accomplished, Planned, and Blockers.
Case 11 (Part. 11)	Social Media Sentiment Agent	Analyze brand mentions; detect sarcasm; calculate sentiment score (-1.0 to 1.0).
Case 12 (Part. 12)	Database Query Assistant	Convert natural language to mock SQL SELECT; guard against DELETE/DROP.

Deep Dive: Case 05 - Order Status Inquiry Agent (Step-by-Step Guide)

1. Business Problem: High volume of 'Where is my order?' queries burden human agents.

2. Workflow Steps:

- *Step 1:* Add Webhook node listening on POST /order-status.
- *Step 2:* Add AI Agent node. Set prompt to ={{ $json.body.message }}.
- *Step 3:* Attach Google Gemini Chat Model subnode (gemini-1.5-flash).
- *Step 4:* Attach Custom Code Tool subnode named getOrderStatus.
- *Step 5:* In tool code, return JSON: { 'ORD-8842': { status: 'Delivered', eta: 'Oct 20' } }.
- *Step 6:* Set System Prompt: 'When user provides an order ID, call getOrderStatus. Be polite and state delivery date.'
- *Step 7:* Test with: 'Where is ORD-8842?' -> Verify agent calls tool and replies accurately.

Module 6 & 7: Vector Databases & RAG Mastery

6.1 High-Dimensional Vector Embeddings Explained

An **Embedding Model** (such as Google's text-embedding-004) maps text strings to 768-dimensional coordinates in geometric space. Sentences with similar semantic meaning end up in close physical proximity, regardless of the specific vocabulary used.

7.1 The Two Lifecycles of Enterprise RAG

Pipeline	Trigger Mechanism	Component Nodes in n8n	Outcome
Pipeline A: Ingestion	Manual / File Upload / Schedule	File Read -> Text Splitter -> Gemini Embeddings -> Pinecone Insert	Knowledge indexed into 768-dim Pinecone vectors.
Pipeline B: Query / QA	Chat Received / Webhook	AI Agent -> Gemini Chat Model -> Pinecone Vector Tool	Semantic chunks retrieved & synthesized into cited answer.

7.2 Trainer Live Demonstration 2: Company Policy RAG System

The trainer demonstrates building both workflows live. Key configuration parameters:

Component	Exact Node ID / Setting	Recommended Value	Rationale
Vector Store	@n8n/n8n-nodes-langchain.vectorStorePinecone	Mode: insert (Ingest) / retrieve-as-tool (QA)	Pinecone serverless manages scaling without infrastructure overhead.
Embeddings	@n8n/n8n-nodes-langchain.embeddingsGoogleGemini	Model: models/text-embedding-004	Outputs 768 dimensions; highly performant and cost-efficient.
Chunking	Recursive Character Text Splitter	Size: 500 chars, Overlap: 50 chars	Prevents breaking sentences across boundary lines.
System Prompt	Policy Grounding Prompt	Must include explicit negative fallback	Forces agent to say 'Not found' rather than hallucinate policies.

ZERO-HALLUCINATION GUARDRAIL

RAG Grounding Guardrail: Always include this exact sentence in your system prompt: 'If the retrieved context does not contain the answer, reply word-for-word: I cannot find this information in the official handbook. Never speculate or extrapolate.'

Part II: Semantic Retrieval (RAG) Assignments (10 Use Cases)

Each participant builds a domain-specific RAG knowledge assistant with dedicated domain documents, metadata schemas, and rigorous factuality testing:

Project Identifier	Domain Assistant	Source Document	Metadata Schema	Key Evaluation Criteria
RAG 01 (Part. 1)	HR Benefits & Leave Advisor	HR_Policy_2026.md	{ section: 'Leave', year: 2026 }	Must state 16 weeks parental leave; abstain on pet insurance.
RAG 02 (Part. 2)	IT Helpdesk Diagnostic Bot	IT_Runbook.md	{ category: 'Network|Hardware' }	Provide exact command to restart print spooler; no guessing on Linux.
RAG 03 (Part. 3)	SaaS API Documentation Bot	API_Docs_v2.md	{ resource: 'Auth|Webhooks' }	Return exact HMAC signature header syntax; cite endpoint URLs.
RAG 04 (Part. 4)	E-Commerce Return Policy Bot	Returns_FAQ.md	{ topic: 'Returns|Shipping' }	Enforce 30-day return window; explain return shipping cost rules.
RAG 05 (Part. 5)	Corporate Travel Policy Advisor	Travel_Rules.md	{ region: 'Domestic|Intl' }	Enforce economy class for flights under 6h; enforce hotel per diem.
RAG 06 (Part. 6)	Warehouse Safety SOP Bot	Warehouse_SOP.md	{ safetyLevel: 'Mandatory' }	Provide step-by-step forklift inspection checklist matching ISO.
RAG 07 (Part. 7)	B2B Sales Battlecard Assistant	Battlecards.md	{ competitor: 'CompX|CompY' }	Pull exact SOC2 Type II audit date and 99.99% SLA credit terms.
RAG 08 (Part. 8)	Microservices Architecture Guide	Arch_Guide.md	{ tech: 'Kafka|Redis|Go' }	Recommend 300ms circuit breaker timeout as written in standard.
RAG 09 (Part. 9)	Legal NDA Compliance Assistant	Standard_NDA.md	{ clause: 'Liability|Law' }	Identify 3-year term of confidentiality and Delaware governing law.
RAG 10 (Part. 10)	New Hire Onboarding Guide	Onboarding_Week1.md	{ phase: 'Day1|Week1' }	Outline exact sequence of Day 1 IT account and email activations.

Objective RAG Quality & Factuality Checklist

Quality Dimension	Verification Technique	Pass Standard	Fail Condition
1. Faithfulness	Compare claims against retrieved Pinecone chunks	100% of facts present in context	Agent states unwritten assumptions
2. Citation	Inspect footer of generated completion	Valid source document and section name	Missing citation or fictional filename
3. Negative Guardrail	Ask question deliberately omitted from document	Replies: 'Information not found in handbook'	Hallucinates plausible but fake policy
4. Chunk Retrieval	Inspect Pinecone node output in execution log	Top-1 chunk contains direct answer	Irrelevant chunks retrieved due to bad query

Module 8, 9 & 10: Advanced Agent Systems, MCP & A2A

8.1 Equipping Agents with Multiple Heterogeneous Tools

Real-world enterprise agents rarely rely on a single tool. A production agent orchestrates multiple heterogeneous tools simultaneously:
• **Tool 1: Pinecone Vector Store** (Queries static internal knowledge bases and handbooks).
• **Tool 2: HTTP Request Tool** (Queries live REST APIs like Stripe, Jira, or Salesforce).
• **Tool 3: Custom Code Tool** (Executes deterministic JavaScript for math, date math, and regex parsing).

8.2 Human-in-the-Loop (HITL) Approval Nodes

For consequential business actions (e.g. issuing refunds > $100, resetting employee passwords, sending customer emails), agents must not execute unchecked. In n8n, we insert an **Approval Node** (e.g. Slack interactive message or Email with Webhook callback). The workflow execution pauses until a manager clicks 'Approve'.

9.1 Model Context Protocol (MCP) in n8n

The Model Context Protocol (MCP) standardizes how AI applications discover and invoke tools. An n8n workflow can act as an MCP server by exposing tools over Webhooks adhering to the JSON-RPC 2.0 specification:

```
// Sample MCP JSON-RPC 2.0 Tool Invocation Payload
{
  "jsonrpc": "2.0",
  "method": "tools/call",
  "params": {
    "name": "query_order_status",
    "arguments": { "orderId": "ORD-9910" }
  },
  "id": "req-001"
}
```

10.1 Agent-to-Agent (A2A) Multi-Agent Collaboration

When business problems require multiple specialist personas, we decouple them into a Coordinator Agent with downstream Specialist Sub-Agents:

Agent Persona	Responsibility	Tool Access	Implementation Node
Coordinator Agent	Deconstructs user request and delegates sub-tasks	AI Agent Tools representing specialist workflows	@n8n/n8n-nodes-langchain.agent
Research Specialist	Performs semantic vector searches and extracts facts	Pinecone Vector Store Tool	Sub-workflow via Execute Workflow
Analytics Specialist	Executes calculations, SQL queries, and validations	PostgreSQL / Custom Code Tool	Sub-workflow via Execute Workflow
Executive Reporter	Compiles final executive briefing with citations	Structured Output Parser / Markdown Formatter	@n8n/n8n-nodes-langchain.agent

Module 11 & 12: Production Readiness & Capstone Evaluation

11.1 Enterprise Production Readiness Checklist

Before promoting any n8n AI Agent to production, verify each audit criterion:

Audit Category	Production Verification Requirement	Failure Impact if Ignored
Security & Credentials	Zero API keys in canvas nodes; all secrets stored in n8n Credential Store.	Severe credential leak risk via canvas exports.
Rate Limit Protection	Retry on Fail enabled with exponential backoff; Wait nodes in batch flows.	HTTP 429 errors crashing production pipelines.
Prompt Injection Guard	User input wrapped in triple quotes; delimiters enforced in prompt.	Malicious users overriding agent instructions.
Human-in-the-Loop	Financial transactions > $100 or password resets require approval node.	Unauthorized automated financial or security actions.
Observability & Alerting	Global Error Trigger workflow sending alerts to Slack/Teams with execution ID.	Silent agent failures unnoticed by operations teams.

12.1 Capstone Project 100-Point Scoring Rubric

Participants demonstrate their end-to-end Capstone against this standardized rubric:

Evaluation Domain	Points	Key Assessment Criteria
1. Business Functionality	15	Solves a real enterprise problem; runs end-to-end without manual intervention.
2. n8n Canvas Architecture	15	Proper separation of concerns; clean subnode wiring (Model, Memory, Tools).
3. Prompt Engineering Quality	15	5-pillar prompt structure with strict constraints and output formatting.
4. Retrieval Accuracy (RAG/Tools)	15	768-dim Pinecone indexing, clean source citations, zero hallucination.
5. Error Handling & Edge Cases	10	Handles empty input, API timeouts, and missing records gracefully.
6. Security & Credential Hygiene	10	All keys vaulted; PII sanitized; prompt injection defenses active.
7. Testing Rigor & Test Suite	10	Documented positive, negative, and adversarial test executions in history.
8. Documentation & Live Demo	10	Clear README, structured 10-minute live demo, articulate Q&A; responses.

Master Facilitator Saturday Intensive Timeline

Turnkey minute-by-minute facilitator timeline for running Session 1:

Time Window	Training Phase	Facilitator Action & Deliverable
09:00 - 09:30	Welcome & Mental Models	Demystify AI hype; contrast deterministic rules vs probabilistic agents.
09:30 - 10:15	n8n Core Mastery	Live build: Webhook -> Transform -> IF -> Respond. Explain $json vs nodeJson.
10:15 - 10:30	Morning Break	Verify any remaining credentials and environment setups for participants.
10:30 - 11:30	Prompt Engineering	5-pillar prompt structure; structured JSON schemas; hands-on classification lab.
11:30 - 12:45	Agent Deep-Dive & Demo 1	Trainer Live Demo 1: Customer Support Triage Agent with Gemini & Custom Tool.
12:45 - 01:45	Lunch Break	Informal discussion on corporate use cases, governance, and architecture.
01:45 - 03:00	Hands-on Lab 1	Participants build assigned Basic Agent from Part I. Trainer assists & debugs.
03:00 - 03:15	Afternoon Break	Prepare Pinecone vector indexes for RAG module.
03:15 - 04:30	Vector DB & RAG Demo 2	Trainer Live Demo 2: Ingestion & QA RAG with Gemini & Pinecone Serverless.
04:30 - 05:15	Hands-on Lab 2	Participants build assigned RAG Knowledge Assistant from Part II.
05:15 - 05:30	Showcase & Capstone Briefing	2 participant demos; review Capstone project specifications and homework.

Master Glossary of Key Terminology

Terminology	Authoritative Engineering Definition
Agentic AI	Autonomous systems capable of goal decomposition, tool selection, and multi-step action execution.
Chunking	Partitioning documents into smaller, cohesive passages with overlap to optimize semantic vector retrieval.
Cosine Similarity	Mathematical metric measuring the directional alignment of two vectors in multi-dimensional space.
Grounding	Anchoring LLM completions in verified, external facts (documents or databases) to prevent hallucinations.
MCP	Model Context Protocol; open standard for exposing tools and resources to AI clients via JSON-RPC.
Pinecone	Managed, cloud-native vector database optimized for low-latency semantic indexing and similarity search.
RAG	Retrieval-Augmented Generation; combining vector search retrieval with generative LLMs for fact-based answers.
ReAct Loop	Reason + Act prompting paradigm where an LLM alternates between reasoning steps and tool executions.
Subnodes	Specialized connection points in n8n cluster nodes providing Models, Memory, Tools, and Parsers.